

View results

Respondent

38

Anonymous

22:48

Time to complete

A. General information about the respondent

1. * Country

Japan



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Permanent Delegation of Japan to UNESCO

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

https://www.unesco.emb-japan.go.jp/itprtop_ja/index.html

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

7. * Full name of officially designated contact person(s) who completed this survey

8. * Position of officially designated contact person(s) who completed this survey

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

11. Brief description of the consultation process established for the preparation of the report

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Japanese National Commission for UNESCO (NatCom) Ministry of Education, Culture, Sports, Science and Technology (MEXT) Cabinet Office (CAO), Ministry of Foreign Affairs (MoFA) National Institute of Science and Technology Policy (NISTEP), National Institute of Informatics (NII), Japan Science and Technology Agency (JST)

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

NatCom <https://www.mext.go.jp/en/unesco/index.htm> MEXT
<https://www.mext.go.jp/en/index.htm> CAO <https://www.cao.go.jp/index-e.html> MoFA
<https://www.mofa.go.jp/index.html> NISTEP <https://www.nistep.go.jp/en/> NII
<https://www.nii.ac.jp/en/> JST <https://www.jst.go.jp/EN/>

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

N/A

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey



Public



Private



Civil Society



Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN
SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?



Yes



No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

Ministry of Education, Culture, Sports, Science and Technology (MEXT) Cabinet Office (CAO),
Ministry of Foreign Affairs (MoFA)

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

- ☒ Yes
- ☐ No

19. 1.2 cont. **If yes**, which language(s)?

Japanese

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

- ☒ Yes
- ☐ No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

The Sixth Basic Plan for Science, Technology and Innovation
<https://www8.cao.go.jp/cstp/kihonkeikaku/6honbun.pdf> In Japanese only

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

23. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

(i)The Principle on Management and Utilization of Publicly Funded Research Data
<https://www8.cao.go.jp/cstp/tyousakai/kokusaiopen/sanko1.pdf> In Japanese only In order to produce excellent research outputs and innovation, this report outlines Japan's basic national approach for the management and utilization of research data based on an open and closed strategy that respects freedom and diversity in research activities while fully taking into consideration both international contributions and national interests. (ii)The Basic Policy on Promoting Open Access to Publicly Funded Scholarly Publications and Scientific Data
https://www8.cao.go.jp/cstp/oa_240216.pdf In Japanese only This report sets out the national policy for realizing the immediate open access to scholarly publications and scientific data based on the following three principles: 1) " Making publicly funded research outouts accessible to the public and contributions to solving global issues ;" 2) Optimizing financial burden of the total amount of subscription fees and article processing charges (APC) ;" 3) " Enhancing the ability to disseminate Japan's research outputs."

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☒ Yes

☐ No

25. 2.1 cont. **If yes**

- please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

Title of the policy: The Sixth Basic Plan for Science, Technology and Innovation the authority in charge of development of the policy entities involved: Cabinet Office the process of development: Cabinet decision year of adoption: March 26, 2021 year of last update: March 26, 2021 key areas/sections of the policy: Science, technology and innovation links to the webpage: <https://www8.cao.go.jp/cstp/english/index.html>

26. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☒ open science infrastructure
- ☐ open engagement of societal actors
- ☐ open dialogue with other knowledge systems

27. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☐ Yes
- ☒ Partly
- ☐ No

28. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

1. The Principle on Management and Utilization of Publicly Funded Research Data
 Title of the policy: The Principle on Management and Utilization of Publicly Funded Research Data
 the authority in charge of development of the policy: Cabinet Office
 entities involved and the process of development: Decision by Integrated Innovation Strategy Promotion Council
 year of adoption: April 27, 2021
 year of last update: April 27, 2021
 key areas/sections of the policy: Science elements of open science that are addressed: Open access to scientific knowledge, open science infrastructure
 links to the webpage(s): https://www8.cao.go.jp/cstp/oa_240216.pdf In Japanese only

2. The Basic Policy on Promoting Open Access to Publicly Funded Scholarly Publications and Scientific Data
 Title of the policy or legal instrument: The Basic Policy on Promoting Open Access to Publicly Funded Scholarly Publications and Scientific Data
 the authority in charge of development of the policy: Cabinet Office
 entities involved and the process of development: Decision by Integrated Innovation Strategy Promotion Council
 year of adoption: February 16, 2024
 year of last update: February 16, 2024
 key areas/sections of the policy: Science elements of open science that are addressed: Open access to scientific knowledge, open science infrastructure
 links to the webpage(s): https://www8.cao.go.jp/cstp/oa_240216.pdf In Japanese only

29. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).



Yes



Under development



No

30. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

Title of the instrument: Open Access Acceleration Project its objective: In anticipation of the mandatory immediate open access to scholarly publications and scientific data on the competitive research funding systems whose new applications will start from fiscal 2025, the project aims at supporting the development of systems for the management and utilization of research results and accelerating the reform information systems and establishment of systems at universities and other institutions for the immediate open access to scholarly publications. The project targets universities and other institutions that have formulated project plans based on a university-wide vision for open access (e.g.open access policy, research data access policy.) the authority in charge: Ministry of Education, Culture, Sports, Science and Technology allocated budget: 10 billion yen (2023) elements of open science included: Response to mandatory open access Title of the instrument: Project for Developing a Research Data Ecosystem for the Promotion of Data-Driven Science its objective: The project aims to develop a research data ecosystem for promoting the sharing and utilization of data, provide functions to support appropriate management of research data, create multidisciplinary and inter-institutional research data search functions, and build a nationwide research data infrastructure that provides support for research data management and human resources development. the authority in charge: MEXT allocated budget: Approximately 1 billion yen per year (2022–2026) elements of open science included: Promotion of sharing and utilization of research data

31. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)



Yes



Under development



No

32. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

the name of ministries and national authorities/entities: Cabinet Office 1. The Principle on Management and Utilization of Publicly Funded Research Data
<https://www8.cao.go.jp/cstp/tyousakai/kokusaiopen/sanko1.pdf> In Japanese only 2. The Basic Policy on Promoting Open Access to Publicly Funded Scholarly Publications and Scientific Data https://www8.cao.go.jp/cstp/oa_240216.pdf In Japanese only

33. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)



Yes



Under development



No

34. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors). To view the elements of open science, see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9>

Each university and institution has established its own policies, such as research data and open access policies, based on the 2021 Recommendations, the Principle on Management and Utilization of Publicly Funded Research Data, and the Basic Policy on Promoting Open Access to Publicly Funded Scholarly Publications and Scientific Data .

35. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ☒ Yes
- ☐ Under development
- ☐ No

36. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

As a funding mechanism for open science, MEXT's Open Science Acceleration Project is providing subsidies to universities and other institutions to assist them in establishing new systems and reforming current information systems for the mandatory implementation of immediate open access. In addition, under its Project for Developing a Research Data Ecosystem for the Promotion of Data-Driven Science, MEXT is providing subsidies to the National Institute of Informatics for the developing and enhancement of a nationwide data infrastructure that promotes data sharing and utilization. https://www.nii.ac.jp/creded/nii_ac_jp_creded.html In Japanese only

37. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☐ Yes

☒ No

38. 2.7 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

39. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☐ Yes

☒ No

40. 2.8 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

41. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

☒ Yes

☐ Under development

☐ No

42. 2.9 cont. **If yes or under development**, please provide details and links as relevant.

Under the Integrated Innovation Strategy, monitoring is being carried out using various indicators, such as the number of institutions that have formulated research data policies and the number of institutional repositories that have been established. [togo2024_zentai.pdf \(cao.go.jp\)](#) p106 In Japanese only

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

43. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

3.65% (2022) Source: Science and Technology Indicators 2024, Research Material-341, 2024, National Institute of Science and Technology Policy, MEXT.

44. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

86.2% (2023) Source: White Paper on Information and Communications 2024, Ministry of Internal Affairs and Communications
<https://www.soumu.go.jp/johotsusintokei/whitepaper/ja/r06/index.html>

45. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

46. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

Science Information NETwork (SINET) is an example of a National Research and Education Network (NREN). SINET is an information and communication network built and operated by the National Institute of Informatics (NII) as science information infrastructure for universities and research institutions throughout Japan. Universities, research institutions. can join the network upon application. SINET6 English TOP - SINET6 - Science Information NETwork 6

47. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.



Yes



Under development



No

48. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☐ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☐ community managed infrastructures, protocols and standards, for example those that support biodiversity and engagement with society
- ☐ platforms for exchanges and co-creation of knowledge between scientists and society
- ☐ community-based monitoring and information systems to complement national, regional and global data and information systems

49. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

The NII Research Data Cloud (NII RDC) is an information platform that supports open science and research integrity and promotes data-driven research. The NII RDC consists of three platforms that match the lifecycle of research data: research data management platform (GakuNin RDM), publishing platform (WEKO3), and discovery platform (CiNii Research).
<https://rcos.nii.ac.jp/en/service/> Commercial/Non-commercial: Non-commercial use

50. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

- ☒ Yes
- ☐ No

51. 3.5. cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

Infrastructure name: Scientific Research Digital Platform The Scientific Research Digital Platform is an academic information infrastructure that promotes the utilization, distribution, and management of research data. It consists of SINET6, a network infrastructure that connects the entire country at ultra-high speeds, and NII RDC (NII Research Data Cloud), a research data infrastructure that enables the management, publication, and search of research data. Date of establishment: April 1, 2022 Language: Mainly Japanese Target groups and beneficiaries: Mainly Researchers Governance/community agreements: Yes Scope: Mainly research institutions, researchers •The Scientific Research Digital Platform supports international initiatives in various fields (e.g., international joint experiments in the fields of high energy physics and nuclear fusion, asteroid probe operations.) •The Scientific Research Digital Platform is intended to be used by researchers in a variety of fields, and aims to promote interdisciplinary collaboration. •The Scientific Research Digital Platform provides broad support from the early stages of research when the researchers themselves manage the research data to the stage of making public and reusing the research data. Related link: <https://www.nii.ac.jp/news/release/2022/0401.html> In Japanese only

52. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☐ Yes

☒ No

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

53. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☒ Yes

☐ No

54. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

The following two activities are carried out by public institutions. •The Gobenrment of Japan (GoJ) is currently examining the standard skills required for research data management personnel, creating a curriculum for training personnel managing research data, and preparing online learning courses, centering on the HRD efforts being made in the Project for Developing a Research Data Ecosystem for the Promotion of Data-Driven Science by the University of Osaka, one of the core institutions in the project. •In order to promote open science, the National Institute of Informatics, which provides online learning courses on research data management through its GakuNin Learning Management System (GakuNin LMS), offers various courses on research data management (for researchers and research assistants.) on its learning management system (website), which provides common online learning content and course records history for higher education institutions. <https://lms.nii.ac.jp/?lang=en> In Japanese only

55. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

56. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

57. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☐ Yes

☒ No

58. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

59. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

☒ Yes

☐ Partly

☐ No

60. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

☒ Yes

☐ No

61. 4.5 cont. **If yes**, please provide more information and links as relevant.

SPARC Japan <https://www.nii.ac.jp/sparc/en/> SPARC Japan conducts advocacy activities related to open access and open science, and collaborates with international initiatives related to open access.

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

62. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☐ Yes

☒ No

63. 5.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

64. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☐ Yes

☒ No

65. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

66. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☐ Yes

☒ No

67. 5.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

68. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☐ Yes

☒ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

69. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

☐ YES, at the national level

☐ YES, at the institutional level

☒ YES, at both the national and institutional levels

☐ No

70. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☒ promotion of pre-prints
- ☐ exploration of open peer-review practices
- ☐ valuing and sharing of negative scientific results and associated data
- ☐ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☐ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☒ promoting open innovation practices
- ☐ Other

71. 6.1. cont. **If yes**, please provide more information and links as relevant

○Jxiv Jxiv is a preprint server in Japan that makes unpublished pre-reviewed papers (preprints) available for open access. It is operated by the Japan Science and Technology Agency (JST) with the aim of supporting the acquisition of pre-emption rights on research results, accelerating the research of preprint submitters themselves by quickly publishing and sharing research papers, as well as revitalizing the entire research community, contributing to global academic research, and building public knowledge. Jxiv enables users to post and publish preprints in all research fields, including the natural sciences, social sciences, and interdisciplinary sciences, in either Japanese or English. Preprints published on Jxiv are assigned a Digital Object Identifier (DOI) and will be released as open access content under a Creative Commons license (CC license). <https://jxiv.jst.go.jp/index.php/jxiv/index> ○NII Research Data Cloud (NII RDC) The NII Research Data Cloud (NII RDC) is an information platform that supports open science and research integrity and promotes data-driven research. The NII RDC consists of three platforms that match the lifecycle of research data: research data management platform (GakuNin RDM), publishing platform (WEKO3), and discovery platform (CiNii Research). <https://rcos.nii.ac.jp/en/service/>

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

72. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

The government shared a common direction on open science and confirmed to cooperate and collaborate to promote it among the countries through the G7 Leaders' Communique and the G7 Science and Technology Ministers' Communique in 2023. As co-chair of the Open Science Working Group (OSWG), a sub-organization under the Science and Technology Ministers' meeting, Japan's members are presenting points of discussion for the future (such as inter-system federation with enhanced interoperability and evaluation indicators that will become international standards) based on international trends in order to promote open science, and promotes international science cooperation by not only calling for international discussion but also sharing information and exchanging views on the implementation status, best practices, and issues of each country's own open science policies. The Government of Japan (GoJ) is considering to support the project for human resource development related to the implementation of recommendations on open science in the Asia-Pacific region to be implemented by UNESCO Jakarta office, utilizing the Japan Fund-in-Trust to UNESCO.

73. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☐ Yes

☒ No

74. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

75. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☐ Yes

☒ No

76. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Part 8. GENERAL CONSIDERATIONS

77. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☒ Yes

☐ No

78. 8.1 cont. If yes, please provide more information.

The distribution of scholarly publications and scientific data is now under the market dominance of global academic publishers, and the subscription fees for academic journals and article processing charges (APC) are increasing considerably. These soaring prices could have adverse effects such as increasing the financial burden on universities, researchers, and others involved in the core activities of academic research, such as subscribing to academic journals and publishing academic papers, and undermining the autonomy of the research community. Moreover, it is necessary to aim establishment and transition to new systems for evaluation and assessment providing incentives for open access while understanding and analyzing the current situation and problems in order to break away the excessive dependence on quantitative indicators in research evaluation and assessment.

79. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org